

Algorithm Aversion

GLOSSARY 02 - SHEET 7 / 8

The tendency to discard an algorithm after seeing it err, even when it outperforms the human

PROFESSIONAL DEFINITION

Algorithm aversion is the tendency of people to lose confidence in an algorithmic forecaster more quickly than in a human forecaster after seeing each make the same mistake, and consequently to prefer the human even when the algorithm is demonstrably more accurate. Its counterpart, algorithm appreciation, is the tendency to over-rely on algorithmic advice in low-stakes or unfamiliar tasks; both are errors of calibration rather than of preference (Dietvorst, Simmons and Massey, 2015; Logg, Minson and Moore, 2019).

WHO DEFINED IT ORIGINALLY

Berkeley J. Dietvorst, Joseph P. Simmons and Cade Massey demonstrated and named the effect at the Wharton School in a 2015 Journal of Experimental Psychology: General paper across five experiments. Jennifer M. Logg, Julia A. Minson and Don A. Moore (2019) established the opposing appreciation effect. The lineage runs to Meehl's (1954) demonstration that statistical prediction outperforms clinical judgement.

WHY THIS DEFINITION WAS DEVELOPED

Meehl's finding had been replicated for six decades without changing practice, which suggested the obstacle was behavioural rather than evidential. Dietvorst and colleagues needed to identify the specific mechanism, and found it in asymmetric updating: an algorithmic error is treated as evidence of a defective system, a human error as an isolated lapse.

FIGURE 1 - ASYMMETRIC RESPONSE TO IDENTICAL ERROR

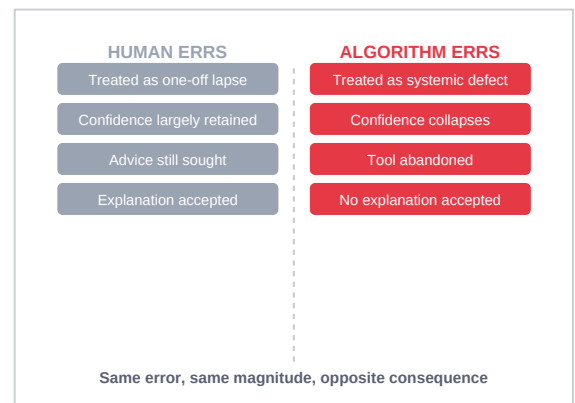


Figure 1. Aversion is an updating asymmetry, not a preference for humans. Granting users even a small right to adjust the model's output substantially restores usage, which makes the remedy a design decision (after Dietvorst, Simmons and Massey, 2015, 2018).

HOW IT IS USED

Anticipate the asymmetry in deployment design: give users limited ability to adjust the model's output, which the same authors showed substantially restores use without materially degrading accuracy, and report model performance against the human baseline rather than against perfection.

WHEN IT IS USED

When a technically superior model is being rejected by its intended users, after a visible model error, when designing the interface through which recommendations are delivered, and when adoption falls after go-live.

BY WHOM IT IS USED

Product owners designing decision-support interfaces, adoption leads diagnosing rejection, model risk functions setting acceptable error tolerances, and managers whose teams have stopped using a deployed tool.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor framing was resistance to change or lack of trust, which treated rejection as a general disposition to be addressed by training and communication. Algorithm aversion identifies a specific, testable mechanism triggered by a specific event - the observed error - and yields a design remedy rather than a persuasion remedy. Dietvorst, Simmons and Massey's (2018) follow-up showing that even trivial adjustment rights restore usage converted a behavioural finding into an interface specification.

RELATED DEFINITIONS IN THIS COURSE

Human-in-the-Loop - Unit 2

AI Literacy - Unit 2

Technology Acceptance Model - Unit 4

Explainability - Unit 6

AI Adoption - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

- Dietvorst, B.J., Simmons, J.P. and Massey, C. (2015) 'Algorithm aversion: people erroneously avoid algorithms after seeing them err', Journal of Experimental Psychology: General, 144(1), pp. 114-126.
- Dietvorst, B.J., Simmons, J.P. and Massey, C. (2018) 'Overcoming algorithm aversion: people will use imperfect algorithms if they can (even slightly) modify them', Management Science, 64(3), pp. 1155-1170.
- Logg, J.M., Minson, J.A. and Moore, D.A. (2019) 'Algorithm appreciation: people prefer algorithmic to human judgment', Organizational Behavior and Human Decision Processes, 151, pp. 90-103.
- Meehl, P.E. (1954) Clinical versus Statistical Prediction. Minneapolis, MN: University of Minnesota Press.